

Circles: Parts, Circumference, and Area (with π)

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Standard: CCSS.MATH.CONTENT.7.G.B.4 — Know and use the formulas for the area and circumference of a circle and use them to solve problems; give an informal derivation of the relationship between the circumference and area of a circle.

Remember from class: The word *circle* comes from the Latin *circulus* = "little ring." $\pi = \frac{C}{D} \approx 3.14$ • Circumference $C = \pi D = 2\pi r$ • Area $A = \pi r^2$ (answer in **square** units). Use $\pi = 3.14$ unless a question says otherwise.

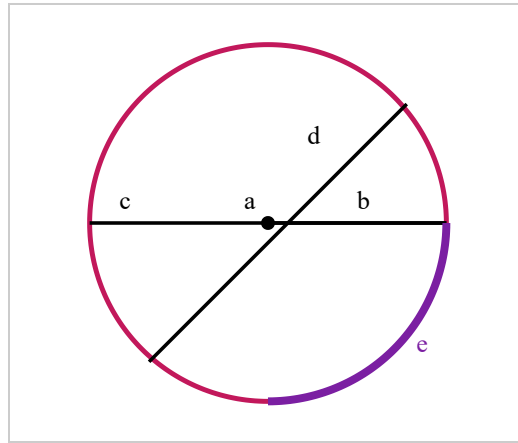
Part A · Parts of a Circle (Vocabulary)

Directions: Use the word bank to answer. Each word is used once.

Word bank: radius • diameter • chord • arc • center • circumference • area

1. The set of all points that are the same distance from the center is the edge of the circle. The straight segment from the center to that edge is the _____.
2. A chord that passes straight through the center, splitting the circle into two equal halves, is the _____. It is exactly _____ times as long as the radius.

3. Match each label to the correct word from the word bank. Write the word on each line.



a, b, c, d, e

a (the dot in the middle) = _____ b (center to edge) = _____
c (edge to edge through middle) = _____ d (slanted edge to edge, not through middle)
= _____
e (the curved purple part of the edge) = _____

4. The total distance *around* the outside of a circle is its _____. The amount of flat space *inside* the circle is its _____.

Part B · Radius ↔ Diameter

Directions: Use $D = 2r$ and $r = \frac{D}{2}$. Show your work.

5. A circle has radius $r = 6$ cm. What is its diameter?

6. A circle has diameter $D = 18$ in. What is its radius?

Part C · Circumference ($C = \pi D = 2\pi r$, use $\pi = 3.14$)

Directions: Find the circumference. Round to two decimal places. Include units.

7. A circle has radius $r = 5$ m. Find the circumference $C = 2\pi r$.

8. A circle has diameter $D = 10$ cm. Find the circumference $C = \pi D$.

9. A round pizza has a diameter of 14 in. How far is it around the crust (the circumference)?

Part D · Area ($A = \pi r^2$, answer in square units)

Directions: Find the area. Use $\pi = 3.14$ unless told otherwise. Square the radius first, then multiply by π .

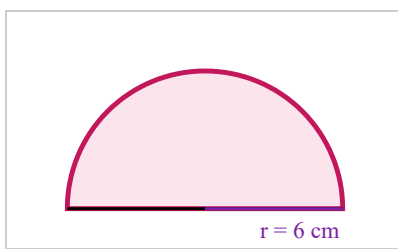
10. A circle has radius $r = 4$ cm. Find the area $A = \pi r^2$ (use $\pi = 3.14$).
11. A circle has diameter $D = 20$ ft. First find r , then find the area (use $\pi = 3.14$).
12. A circle has radius $r = 7$ cm. Find the area using the fraction $\pi = \frac{22}{7}$. (Hint: this r makes the 7 cancel nicely!)

13. A circle has diameter $D = 28$ m. Find r , then find the area using $\pi = \frac{22}{7}$.

Part E · Half & Quarter Circles

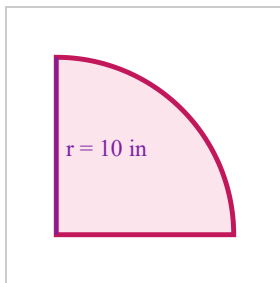
Directions: Find the area of the shaded region. A half circle is $\frac{1}{2}$ of a full circle's area; a quarter circle is $\frac{1}{4}$. Use $\pi = 3.14$.

14. Find the area of this **half circle** (radius $r = 6$ cm).



half circle

15. Find the area of this **quarter circle** (radius $r = 10$ in).



quarter circle

Part F · Explain in Words (Think First!)

Directions: Write full sentences. For these, your **reasoning** matters as much as the number.

16. The circumference of a circular garden is $C = 6.28$ m.

(a) In words, explain how you can use the circumference to find the radius. Then find r .

(b) Now find the area of the garden. Show how you got it.

(c) **Bonus — explain before formulas:** If you *doubled* the radius of this garden, what would happen to the circumference? What would happen to the area? Explain *in your own words first* (without numbers), then check with the formulas.